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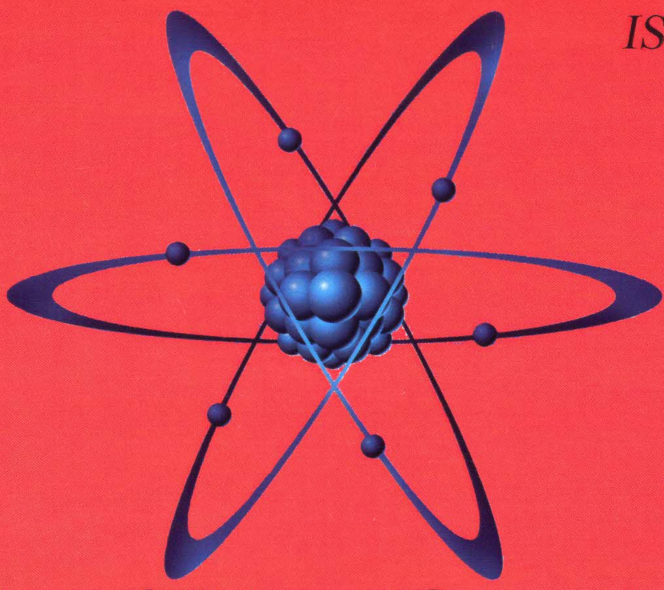

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FINANCIAL, MATERIAL AND INFORMATION FLOWS OF THE COMPANY

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Abstract. *The article deals with the issues of varied connections of big, medium and small business companies. The possible variants of breaking these connections are shown to which we refer fragile, plastic and super plastic. The definitions of these connections are given.*

In the works [3, 4] we have described material, financial and information flows that connect the corporation with as other companies and firms so as with itself. Let us study these flows and their characteristics and also possible variants of the connections and their changes at the economic decrease.

So, in any company one can define the following types of the flows (TF): material (MT), financial (FN) and information (IN), that is they can be represented as a sum (1),

$$TF_d = \{MT_1, \dots, MT_h, FN_1, \dots, FN_j, IN_1, \dots, IN_w\}. \qquad (1)$$

To the material flows we refer the flows with the help of which the product itself is made that is produced by the corporation so as the energy resources which are needed for its make.

Thus, to the basic material and energy flows we can refer the following: devices (DV); units (UN); details (DT); semi-finished product (SF); workpieces (WP); cutting attachments (CA); measuring tools (MT); equipment (EQ); oil, petrol, oils, liquid coolant, etc. (OM); coal (CL); gas (GS); electricity (EL).

It can be represented by the expression (2),

$$MT_h = \{DV_1, \dots, DV_s, UN_1, \dots, UN_j, DT_1, \dots, DT_w, SF_1, \dots, SF_r, WP_1, \dots, WP_u, CA_1, \dots, CA_o, MT_1, \dots, MT_p, EQ_1, \dots, EQ_a, OM_1, \dots, OM_f, CL_1, \dots, CL_k, GS_1, \dots, GS_z, EL_1, \dots, EL_c\}. \qquad (2)$$

To the financial flows one should refer the flows that are connected directly with money and we refer to them the following:

- money payments using the no-cash bank system both as commodity turnover so as non-commodity turnover (BMP);
- money payments to employees by transferring the money to magnet cards and by cash (MPE);
- bills of credit and other securities (S).

The financial flows can be presented by the following combination (3),

$$FN_j = \{ BMP_1, \ldots, BMP_h, MPE_1, \ldots, MPE_j, S_1, \ldots, S_w \}. \quad (3)$$

To the information flows one should refer the flows that are connected with mass media (MM), electronic networks (wired and wireless) and also at personal communication and at overhearing or eavesdropping. Information is the most important means of managing. From the point of view of cybernetics, managing is a process of the purposeful information processing. Information is as work object so as work product in management [5]. One of the most important varieties of information is *economic* information and its distinctive feature is its connection with the managing processes of peoples' masses and companies. The economic information accompanies the production processes, distribution, exchange and consumption of the material goods and services where its significant part is connected with the social production.

The *economic* information (EI) is a sum of data that appears in the industrial and economic activity (DIEA), commercial (DCA) and financial activity (DFA) and which are used to perform the functions of the organizational-economic management of this activity (4),

$$EI_d = \{DIEA_1, \ldots, DIEA_h, DCA_1, \ldots, DCA_j, DFA_1, \ldots, DFA_w\}. \quad (4)$$

At this the economic information is characterized by the following parameters: big volume (BV); multiple repetition of receiving and processing cycles (MRRPC); time regulations of information process procedures (TIP); significant specific weight of the Telecommunication procedures and logic operations of the information processing (SSWPO); comparatively simple calculations for most information types (CSCI) [1] that is it can be represented as the following function (5),

$$EI_u = F(BV_h, MRRPC_j, TIP_w, SSWPO_f, CSCI_e). \quad (5)$$

The necessary information can be received by different ways such as:

- journals, magazines, newspapers, booklets (periodicals) (JNB);
- books, advertising pamphlets (BAP);
- billboards (BB);
- information received by networks in the electronic way (telephone, cable, optical fiber) (IRN);
- information received by the TV (IRTV);
- information received by the radio (IRR);
- information received by the cellular communications (wired and satellite phone) (TEL);
- information received at personal communication, overheard or eavesdropped (IPOE).

Thus, the ways of receiving the information (WRI) can be written as the following expression (6),

$$WRI_h = \{ JNB_1, \ldots, JNB_s, BAP_1, \ldots, BAP_j, BB_1, \ldots, BB_w, IRN_1, \ldots, IRN_r, IRTV_1, \ldots, IRTV_u, IRR_1, \ldots, IRR_o, TEL_1, \ldots, TEL_p, IPOE_1, \ldots, IPOE_a\}. \quad (6)$$

All mentioned above connections can be subdivided into three classes: fragile, plastic and super plastic.

In such a way we can give the definition of the existing connections of any company.

The fragile connections are called those that are broken even at slight changes of the economic condition of the company in the direction of the made products reduction or services.

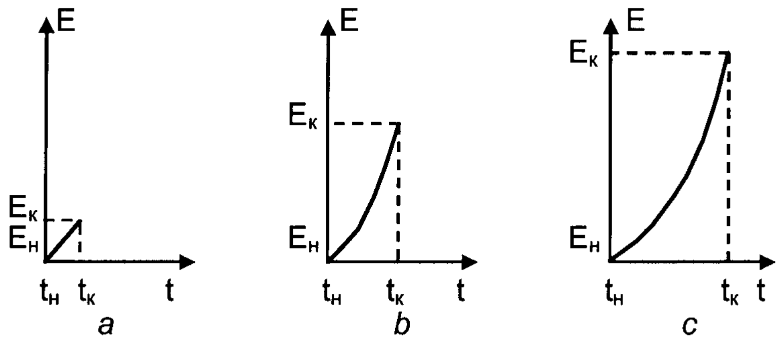
So, in this case we can observe a complete closing of the production goods and services that can be referred to the secondary and which are not the basic product. One can refer here such products and services the production of which has been reducing lately because of its low demand or it has been planned to reduce or close it. Thus, worsening of the economic situation as of the company so as the economic situation in general promotes the acceleration of making decisions by the company's heads or the liquidation of such subdivisions, or their reorientation taking into consideration the market opportunities change. It should be noted that to the fragile connections of the company we refer the connections of the company with other ones that give their products or services and which economic position leaves much to be desired. Hence, breaking of the company's connections can be initiated by the company itself so as by the company with which there has been that connection.

The plastic connections of the firm are those that are broken at significant changes of the economic condition of the company in direction of the reduction of produce or services.

The superplastic connections of the company are those connections that can be broken only when there is an economic collapse or the company liquidation.

The types of the fragile connections can be presented as picture 1*a*.

In picture 1 we can see that the fragile connections are broken very fast even at slight changes of the economy condition. Here one can introduce the following value ΔE as the difference between the initial E_i and final E_f condition of the economic recession. If the economy of the company (country) is at its boost then that value ΔE is positive. At the economy decline it is correspondingly has a negative meaning.



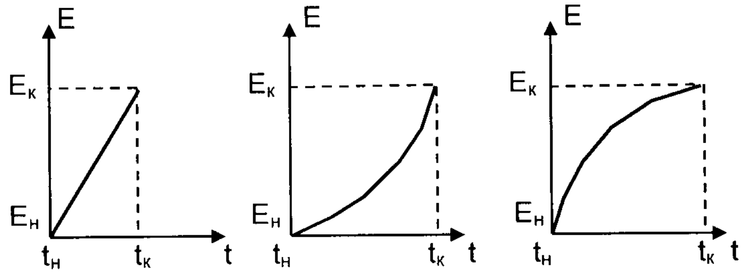
Pic. 1 Types of the company's connections at the economic decline
a) fragile b) plastic c) superplastic
where, t- time of the beginning of the economic decline, $t_к$ - time of the end of the economic decline, E_H- the economy condition at the beginning of the economic decline, $E_к$- the economy condition at the end of the economic decline

To give a more detailed description of the changed economy it is better to use the value of the relative change in percentage that we will calculate by the following formulae (7),

$$\Delta E = \frac{E_к - E_н}{E_к} \cdot 100. \tag{7}$$

We will assume that at the values: 0% ≤ ΔE ≤ 10% the fragile breaking of the connections takes place, picture 1*a*; at 0% ≤ ΔE ≤ 90% – the plastic one, picture 1*б*, and at 0% ≤ ΔE ≤ 100% – the superplastic breaking takes place, picture 1*в*. That is, the superplastic connections can be broken between the companies only at the economic collapse. For example, a heavy earthquake, tsunami or other natural cataclysms for a small state can lead to its collapse or its crash.

Here we can introduce the following value $\Delta t = t_e - t_b$ that shows the time interval during which the company's economic decline takes place. One should take into consideration the function kind ΔE = F(Δt) according to which this change takes place. In picture 2 it is shown three possible variant of the company's connections break. For the calculations and prediction of the company's economic condition change it is reasonable that these changes should go by the linear dependence that will allow easy predicting the time duration basing on the previous experience and having the statistic data as a data base.

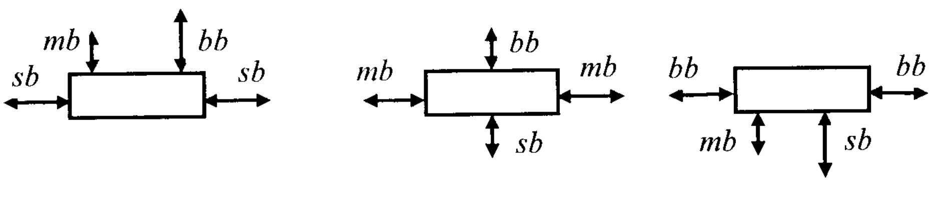


a b c

Pic. 2 Variants of developing the connections break
a – Linear, b – Exponential, c - Polynominal

To the superplastic connections we can refer the information connections, the mass media at the first rate, excluding the internet because the small business can refuse it, or even not use it at all if the company consists of a small number of employees. Here one should refer the material flows without which the industry ill stop and also the money flows, the energy flows, etc.

The superplastic connections can be divided into two classes: main and secondary. To the main connections we refer the connections that have existed since the moment of the company's legal liquidation. To the secondary ones we refer the connections that exist after the company's legal liquidation. That is, the former employees of the firm, especially if they have been connected by kindred bounds, will naturally communicate between each other and if the economic situation becomes better they will be able to renovate their firm to produce this or that product or services again. It, at the first rate, is about the small business and also medium business with a small number of employees.



a b c

Pic. 3 Types of the company's connections
a - Small business (sb), b - medium business (mb), c - big business (bb)

All other connections refer to the plastic ones and are an intermediate condition between the fragile and superplastic connections.

In his works [3, 4] the author has shown that the economy of any state can be represented as coats, hence the company's connections can be drawn as in picture 3. As we see in picture 3 the small business has the following types of connections: the connection with small business, with medium business and connection with big one. The number of connections of the small business can vary from 1 to several tens. The medium business has connections with small busines, medium business and big business and the number of these connections can vary from several tens to several hundreds. The big business is connected with a great number of suppliers of different godds and serveices, as with small, medium , so as with big business correspondingly. The number of the connections can vary from several hundreds to several ten thousands and more.

The connections break can be interpreted as endless stem stretch and this law is used at unlimited growing tension [2]. Here one should take into consideration the problem of the influence on the breaking time the creep law. That is, at worsening the economic condition of the company the connections between other companies can break as immediately so as gradually depending on many reasons such as, for example, time of the decline process, decline intensity, the connections condition at the current moment and many other. And at the moment of time t_f the final break of the connections takes place.

Thus, one can draw the following conclusion that if the company has a daily detailed statistic account of the company's condition so as its connections and it observes the market condition in the way of the data base and has a good base of knowledge it can predict the coming of the crisis situation and can take measures to reduce to the minimum the influence of the economic recession on its condition.

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